

Senior Applied AI Engineer Project Challenge Guide

This round evaluates two things: your ability to build a practical AI solution and explain your decisions clearly. For this role, implementation depth matters most.

What you should build

Build a proof-of-concept agent that runs locally on the provided synthetic data and solves one common incident type well. It should work end to end and reflect strong engineering judgment.

Your solution should show the full flow:

- ingest a ticket
- retrieve the right knowledge or asset context
- generate grounded step-by-step guidance
- produce a clear summary
- handle at least one edge case
- include at least one guardrail such as escalation, grounding, or output validation

A working live demo is mandatory.

Deliverables

Please come prepared with:

- a presentation deck covering the problem, pilot scope, solution, evaluation, and scale-up plan
- working code for your proof of concept
- a live end-to-end demo
- architecture diagrams
- a clear README or setup guide
- evaluation results with metrics, especially routing accuracy and knowledge-base retrieval accuracy

Presentation guidance

Aim to finish your presentation and demo in about 30-40 minutes, leaving 20-30 minutes for discussion.

Your presentation should cover:

- the business problem
- the pilot scope, deliverables, and success criteria

- the solution and key technical decisions
- how the solution can scale beyond the pilot

What the panel will evaluate

Be ready to go deep on:

- your RAG and retrieval pipeline
- chunking, indexing, embeddings, vector store, thresholds, and reranking
- agent orchestration and grounding
- evaluation methodology and results
- code quality and engineering decisions

Use real evaluation numbers, not only qualitative claims.

How to present

Keep the business story simple and clear, but be ready for detailed technical questions on implementation choices, failure modes, guardrails, and trade-offs.

Common mistakes to avoid

- no working demo
- a demo you cannot defend technically
- vague evaluation
- messy or undocumented code
- a solution that only works on a rehearsed example

Final advice

Approach this like a senior engineer building an AI system for real-world use. Show thoughtful design, honest evaluation, clear limitations, and strong technical judgment.